

Do fast foods cause asthma, rhinoconjunctivitis and eczema? Global findings from the International Study of Asthma and Allergies in Childhood (ISAA...

BACKGROUND: Certain foods may increase or decrease the risk of developing asthma, rhinoconjunctivitis and eczema. We explored the impact of the intake of types of food on these diseases in Phase Three of the International Study of Asthma and Allergies in Childhood. **METHODS:** Written questionnaires on the symptom prevalence of asthma, rhinoconjunctivitis and eczema and types and frequency of food intake over the past 12 months were completed by 13-14-year-old adolescents and by the parents/guardians of 6-7-year-old children. Prevalence ORs were estimated using logistic regression, adjusting for confounders, and using a random (mixed) effects model. **RESULTS:** For adolescents and children, a potential protective effect on severe asthma was associated with consumption of fruit ≥ 3 times per week (OR 0.89, 95% CI 0.82 to 0.97; OR 0.86, 95% CI 0.76 to 0.97, respectively). An increased risk of severe asthma in adolescents and children was associated with the consumption of fast food ≥ 3 times per week (OR 1.39, 95% CI 1.30 to 1.49; OR 1.27, 95% CI 1.13 to 1.42, respectively), as well as an increased risk of severe rhinoconjunctivitis and severe eczema. Similar patterns for both ages were observed for regional analyses, and were consistent with gender and affluence categories and with current symptoms of all three conditions. **CONCLUSIONS:** If the association between fast foods and the symptom prevalence of asthma, rhinoconjunctivitis and eczema is causal, then the findings have major public health significance owing to the rising consumption of fast foods globally.